

MAC 201 Course Outline

MAC 201 — Macroeconomic Frameworks

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	30 (indicative band 24–32)
Shape	5 modules · 10 sessions · 6 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101, or none. This is the first course of School 02 and it assumes no macroeconomics beyond an undergraduate introduction. No modelling is required and nothing is run: every input is a published CSV and every laboratory can be done in a spreadsheet. It should be taken before MAC 202, and before any of the PFM courses.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-25 · 12-month cycle

What this course is

MAC 201 is the first course of School 02 and it has no engine. It takes the twelve accounting identities of §8.1 of the Academy command, states each one properly, and then does the thing nobody does: evaluates every one of them against published statistics for one country.

The sources are the World Bank’s World Development Indicators and six IMF statistical families, for Utopia over 2000–2024. Ten of the twelve identities can be evaluated. 5 of the tests do not close, 2 produce two answers, and the one that closes exactly turns out to be the least informative result in the course, because the deflator it tests is defined as a ratio rather than measured.

The one habit this course exists to install. Before you quote a macroeconomic number, say which concept, which source and which denominator.

Who it is for

Anyone who has to read or produce a macroeconomic figure: budget and planning staff, macro-fiscal units, analysts in central banks and oversight bodies, researchers using published statistics, and jour-

nalists or advisers who quote them. It is also the recommended entry point for anybody starting the modelling courses in School 03.

What this course does not claim

This course is one country and one vintage. Every number in it is Utopia, extracted on 2026-08-25. The identities are general; the residuals are not, and another country would produce different residuals and in some cases different signs. The final assignment exists to make that concrete.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	What a macroeconomic framework is	01–02	01–02	5.2
M2	The four sectors	03–05	01, 03	7.8
M3	Testing the spine against data	06–08	01, 04–05	8.8
M4	The findings	09	05	2.6
M5	Reading and reporting	10	06	2.6
ASM	Assessment and revision	—	—	3.0
	Total	10	6	30.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

M1 What a macroeconomic framework is — Sessions 01–02 · 5.2 h · Laboratory 01–02

- **01 What a macroeconomic framework is** — the question this course exists to answer; four sectors, one economy; what an identity is, and what it is not; the twelve identities of the spine, and more
- *Lab 01 Close the expenditure identity, and fail* — establish for yourself that the identity every model starts from does not hold in published data, and show that the residual you computed is the one the compiler published
- **02 Units, sources and the three questions** — units, prices and volumes; sources, vintages and benchmarks; the three questions to ask of any published figure
- *Lab 02 Find the break* — detect a benchmark revision in a series that looks continuous, and establish the window over which a ratio may legitimately be computed

M2 The four sectors — Sessions 03–05 · 7.8 h · Laboratory 01, 03

- **03 The real sector** — the real sector: what GDP is; the three approaches, and why they disagree; reading the expenditure accounts; current prices, constant prices and chain-linking
- **04 The fiscal sector** — the fiscal sector: what a government account records; revenue: taxes, non-tax and grants; expense, expenditure, and the acquisition of assets; the balances: operating..., and more
- *Lab 03 Build the fiscal balance four ways* — establish that ‘the deficit’ is not identified without a sector, a set of terms and a compiler
- **05 The external and monetary sectors** — the external sector: residency, not nationality; the current account and the rest of the world; units and the exchange rate; the monetary sector: what money is a liability of, and more

M3 Testing the spine against data — Sessions 06–08 · 8.8 h · Laboratory 01, 04–05

- **06 Testing an identity** — method: how to test an identity against data; I-01 — the identity that closes, and why that is uninformative; I-03 — the expenditure identity, and ten per cent of GDP; I-03 again — three...
- **07 The fiscal and debt tests** — I-06 — six sector codes, six deficits; I-07 — the stock-flow adjustment
- *Lab 04 Measure the stock-flow adjustment* — compute the residual of the debt identity, name it, and state what a projection that ignores it is assuming
- **08 Prices, denominators and what cannot be tested** — I-08 — one stock, two ratios; I-11 — two inflations; the three identities that cannot be evaluated at all
- *Lab 05 Which inflation? Which denominator?* — show that two correct answers to the same question can differ by points, and that a ratio is not defined until its denominator is

M4 The findings — Session 09 · 2.6 h · Laboratory 05

- **09 The seven findings** — what makes a finding a finding here; F-01 — The expenditure identity does not close, by ten per cent of GDP; F-02 — The same identity, the same year, three different..., and more

M5 Reading and reporting — Session 10 · 2.6 h · Laboratory 06

- **10 Reading, reporting and handing on** — what this course does not claim; writing a country note from published statistics; reviewing someone else's macroeconomic figures; running this for another country, and more
- *Lab 06 Write the country note* — produce two pages on Utopia's macroeconomic position in which every figure carries a concept, a source and a denominator — and in which the last paragraph is about what is not known

What a graduate can do

1. Name the four sectors of a macroeconomic framework and state what each records.
2. State what an identity is, and why published data cannot refute one.
3. Explain what has happened when a measured identity does not close.
4. Distinguish definitional, behavioural and equilibrium relations.
5. State the twelve identities of the accounting spine.
6. Name the four kinds of residual and identify which one a given residual is.
7. Settle currency, price basis, scale and fiscal-year basis before reading any figure.
8. Explain why chain-linked volume components are not additive.
9. Detect a benchmark revision from a series alone, and set a comparable window.
10. Apply the three questions — concept, source, denominator — to any published figure and rewrite the sentence.

The 16 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	86 items across 10 modules

Component	Weight	What it is
Laboratory exercises	30 %	6 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	the identity that fails, and the break that hides; the fiscal balance, four ways; the residual with a name, and the two-answer questions; the country note, and another country
Final assessment	30 %	Three hours, open book: definitions and structure; calculation; interpretation; reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A 118-page technical manual; 10 lecture decks and 6 laboratory decks; a laboratory manual, instructor guide and learner workbook; a 86-item quiz bank, 4 assignments and a final assessment with its marking guide; 12 datasets; 5 code exercises; 6 case studies; and the lms package. The README lists every file.

Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

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